

SAJAN KUMAR

Ramgarh Panchkula Haryana

☎ +918873078315 ✉ kumarsajan0013@gmail.com 💼 linkedin.com/in/sajan-kumar-singh 🐙 github.com/SajanKrSingh

Education

Chandigarh University

Master of Computer Applications (MCA)

Aug. 2023 – May 2025

Mohali, Punjab

T P Verma College

Bachelor of Computer Applications (BCA)

Aug. 2019 – May 2023

Narkatiyaganj, Bihar

Technical Skills

Languages: JavaScript (ES6+), Java, Python, SQL, HTML5, CSS3

Frontend: Next.js 15 (App Router), React.js, Tailwind CSS, Redux, Bootstrap, Framer Motion

Backend: Node.js, Express.js, FastAPI, Firebase (Auth/Firestore), MongoDB, WebSockets

Tools & DevOps: Docker, Render, Git, GitHub, Postman, Razorpay API, NodeMailer, Murf.ai

Experience

EdiGlobe

Web Developer (Next.js & MERN Stack)

Jan 2025 – Present

Bengaluru

Overview: Developed a full-stack Enterprise CRM HRMS solution to automate the entire lifecycle of the business. The system replaces manual Google Forms with a custom **"Lead Gen Engine"** and streamlines operations using **Next.js 15, Firebase, and Tailwind CSS**.

- **1. Custom Lead Generation Engine (Google Form Alternative)**
 - Built a dynamic system to generate public student registration pages with professional UI (SVG animations, Glassmorphism), replacing static Google Forms.
 - Implemented **Smart Auto-Assignment Logic** in the backend that automatically routes incoming leads to the specific Sales Executive who shared the link, eliminating manual distribution.
 - Designed real-time processing triggers that send instant "Thank You" emails to candidates and update the Sales Dashboard via Webhooks.
- **2. Advanced Authentication Onboarding System**
 - Solved the challenge of onboarding existing staff by building a **Superadmin Tool** for manual account creation within Firebase Auth.
 - Automated the **Credential Delivery System** using NodeMailer; the system auto-generates secure passwords and emails a "Welcome Kit" containing login credentials and a unique Digital ID Card.
 - Engineered strict **Role-Based Access Control (RBAC)** middleware to separate dashboards for Superadmin, HR, Sales, and Candidates.
- **3. HR Employee Management Module**
 - Developed a **Digital Attendance System** using dynamic QR Codes; employees possess a unique QR on their profile which HR scans to mark "Check-In/Out" instantly.
 - Created a centralized Employee Repository for managing sensitive documents (Offer letters, IDs) and tracking performance statistics.
- **4. Sales CRM Financial Integration**
 - Integrated **Razorpay API** to build a custom Payment Link Generator, enabling sales teams to create partial/full payment links directly from the CRM.
 - Designed a Kanban-style **Lead Pipeline** (New → Converted) with "Bulk Transfer" capabilities to reassign leads between executives.
 - Built an Analytics Command Center using **Recharts** to visualize monthly revenue, attendance trends, and sales funnels for upper management.

Projects

Zarex - Real-time AI Voice Assistant | *Python, FastAPI, Docker, Murf.ai*

December 2025

- Developed a multi-skilled, cloud-native conversational AI assistant, integrating **Murf AI (TTS)**, **Google Gemini (LLM)**, and **AssemblyAI (STT)**.
- Solved audio latency issues by switching from HTTP to **WebSockets** and implementing **AudioWorkletNode API** for glitch-free, real-time audio streaming.
- Designed a modular backend architecture to manage multiple APIs and created a dynamic UI modal for secure API key input (removing .env dependency).
- Implemented switchable AI Personas (Assistant, Tutor, Comedian) and multi-skill integration for live web search and weather updates.
- Containerized the application using **Docker** to resolve dependency conflicts and successfully deployed the production build on **Render**.

DestinPath.AI (Travel Planner) | *React.js, Firebase, Gemini AI*

December 2024

- Designed an AI-driven travel itinerary generator integrating **Google Gemini API** and Google Places API to create fully personalized trip plans.
- Implemented complex prompt engineering to parse user budgets, location, and travel preferences into structured, day-by-day travel schedules.
- Utilized **Firebase Authentication** for secure login and Cloud Firestore to save and sync user itineraries across devices in real-time.
- Achieved a 90% reduction in trip planning time for users through automated content generation and instant hotel recommendations.

Smart Learning Management System | *MERN Stack, Redux, Razorpay*

September 2024

- Architected a scalable LMS platform with distinct dashboards for Instructors (course creation/analytics) and Students (consumption/progress).
- Integrated **Razorpay Payment Gateway** with webhooks to handle secure transactions and automatically enroll students upon successful payment.
- Implemented secure file handling using Multer for video/document uploads and built progress tracking algorithms to monitor course completion.
- Designed a responsive frontend with **Redux Toolkit** for global state management, reducing redundant API calls by 30%.

Hospital Management API | *Node.js, Express.js, MongoDB*

July 2024

- Developed a robust RESTful API backend handling complex relationships between Doctors, Patients, and Medical Reports.
- Implemented **JWT (JSON Web Tokens)** for stateless authentication and bcrypt for password hashing to ensure data security.
- Designed scalable database schemas with MongoDB indexing to support high-volume request handling for patient records.

Certifications

30 Days of AI Voice Agents Challenge – Murf.ai (Completed)

December 2025

Built 'Zarex', a cloud-native voice assistant using FastAPI, Docker, and WebSockets.

Full Stack Web Development – PW Skills

Data Structures Algorithms in Java – Coding Ninjas

Frontend Backend Development – Coding Ninjas